Project Report: Restaurant Recommendation Clustering and Classification

1. Introduction

This project focuses on analyzing food delivery data scraped from the Swiggy website to classify restaurants based on various features such as food type, rating, delivery time, and location. Due to the absence of an explicit target column for classification, unsupervised learning was initially applied to form meaningful clusters, which were later used for training supervised models. The overall objective was to cluster restaurants into recommendation groups and then build a classification model to predict cluster membership for new records.

2. Dataset Description

The dataset was scraped from Swiggy and consists of the following features:

- Hotel Name
- Rating
- Delivery Time (in minutes)
- Food Type
- Location
- Offer Above Price
- Offer in Percentage

The original data was uncleaned and required extensive preprocessing before model development.

3. Exploratory Data Analysis (EDA) Key Points

- The dataset contains fewer than 2,000 records, which may increase the risk of underfitting or overfitting.
- Although there are no null values, many fields are marked as "Not Available," especially in optional fields.
- All features were originally of object data type, which restricted direct statistical analysis.
- Most values in each feature are unique, limiting traditional aggregation techniques.
- There were 19 duplicate rows that needed removal.
- The 'Food Type' feature was expected to cause increased dimensionality after encoding due to its diverse entries.

4. Data Preprocessing Summary

1. Dropped the 'hotel_name' column as it did not contribute to the prediction task.
2. Cleaned the 'rating' column by handling inconsistent patterns and filling missing values with the median.
3. Processed 'time_minutes' by handling range-like values and replacing missing data with the median.
4. Label encoded the 'location' feature to convert it to numerical format.
5. Replaced all occurrences of 'not_available' with NaN in 'offer_percentage'.
6. Dropped 'offer_percentage' and 'offer_above' due to the dominance of missing data.
7. Encoded 'food_type' using frequency encoding to preserve information without increasing dimensionality.
8. Removed all duplicate records to ensure data integrity.
9. Applied standard scaling to the full dataset.
10. Used robust scaling to reduce the influence of outliers, which was ultimately ineffective.
11. Applied power transformation to 'food_type', 'location', and 'rating' to address skewness in distribution, yielding improved results.

5. Clustering Approach

Since no target label was initially available, KMeans clustering was applied to generate artificial class labels. While the initial hypothesis was to split the data into 2 clusters, further analysis using Elbow and Silhouette plots showed that 3 clusters provided a more balanced distribution, better performance metrics, and clearer separation.

Final Number of Clusters: 3

This decision was supported by:

- A noticeable elbow in the Elbow Plot at k=3
- Higher Silhouette Score for 3 clusters compared to 2
- Better balance in class distribution

6. Model Selection and Evaluation
Multiple classification algorithms were trained using the cluster labels as the target. The models included:

- Logistic Regression
- Random Forest
- Gradient Boosting
- Decision Tree

Best Performing Model: Logistic Regression

- Accuracy: 0.9845
- Precision: 0.9847
- Recall: 0.9845
- F1-Score: 0.9845
- ROC-AUC: 0.9997

Model Selection Rationale:
Logistic Regression performed the best across all metrics, closely followed by Decision Trees. Both models showed strong performance in terms of precision, recall, and F1-score, and demonstrated minimal variance between training and validation scores, indicating good generalization.

7. Deployment

The project has been deployed using **Streamlit**, enabling real-time interaction through a user-friendly web interface. The application collects user inputs such as rating, delivery time (in minutes), food type, and location. Although optional fields like hotel name, offer above price, and offer percentage are included, only the relevant fields are processed.

The app applies data preprocessing steps such as encoding categorical features (food_type, location) and scaling (StandardScaler, RobustScaler, PowerTransformer) directly within the interface. After transformation, the cleaned input is passed to a pre-trained **Logistic Regression** model to predict the appropriate recommendation cluster (0, 1, 2). The predicted cluster is then displayed to the user in an organized and interactive format